

TASK 6 — Tags and Test Execution

1. Experiment Title

Tags and Test Execution Using Robot Framework

2. Problem Statement

Implement test cases in Robot Framework using **tags** to categorize tests and demonstrate selective test execution using `--include` and `--exclude` options. Also execute the test cases in parallel using multiple processes.

3. Objective

- To understand and implement **tags** in Robot Framework.
- To categorize test cases using tags such as smoke and regression.
- To execute selected tests using `--include`.
- To exclude specific tests using `--exclude`.
- To execute test cases in parallel using multiple processes.

4. Tools, Software and Concepts Used

Tools & Software

- Python
- Robot Framework
- SeleniumLibrary
- Selenium WebDriver
- Google Chrome
- VS Code
- Robot Framework Pabot

Concepts Used

- Robot Framework Test Cases
- Test Tags
- [Tags]

- --include
- --exclude
- Parallel Test Execution
- pabot
- Selenium Web Automation
- Explicit Waits

5. Source Code / Implementation Steps

File Name

assignment6.robot

Source Code

*** Settings ***

Library SeleniumLibrary

*** Test Cases ***

Valid Login Test

[Tags] login smoke

Open Browser https://www.saucedemo.com/ Chrome

Wait Until Element Is Visible id=user-name 10s

Input Text id=user-name standard_user

Input Text id=password secret_sauce

Click Button id=login-button

Location Should Contain inventory.html

Close Browser

Invalid Login Test

[Tags] login regression

Open Browser https://www.saucedemo.com/ Chrome

Wait Until Element Is Visible id=user-name 10s

Input Text id=user-name wrong_user

Input Text id=password wrong_password

Click Button id=login-button

Wait Until Element Is Visible css=h3[data-test="error"] 10s

Page Should Contain Element css=h3[data-test="error"]

Close Browser

Homepage Test

[Tags] smoke

Open Browser https://www.saucedemo.com/ Chrome

Wait Until Element Is Visible id=user-name 10s

Page Should Contain Element id=user-name

Page Should Contain Element id=password

Close Browser

Execution Commands

Run all test cases:

robot assignment6.robot

Run only smoke tests:

robot --include smoke assignment6.robot

Run only regression tests:

robot --include regression assignment6.robot

Exclude smoke tests:

```
robot --exclude smoke assignment6.robot
```

Install Pabot for parallel execution:

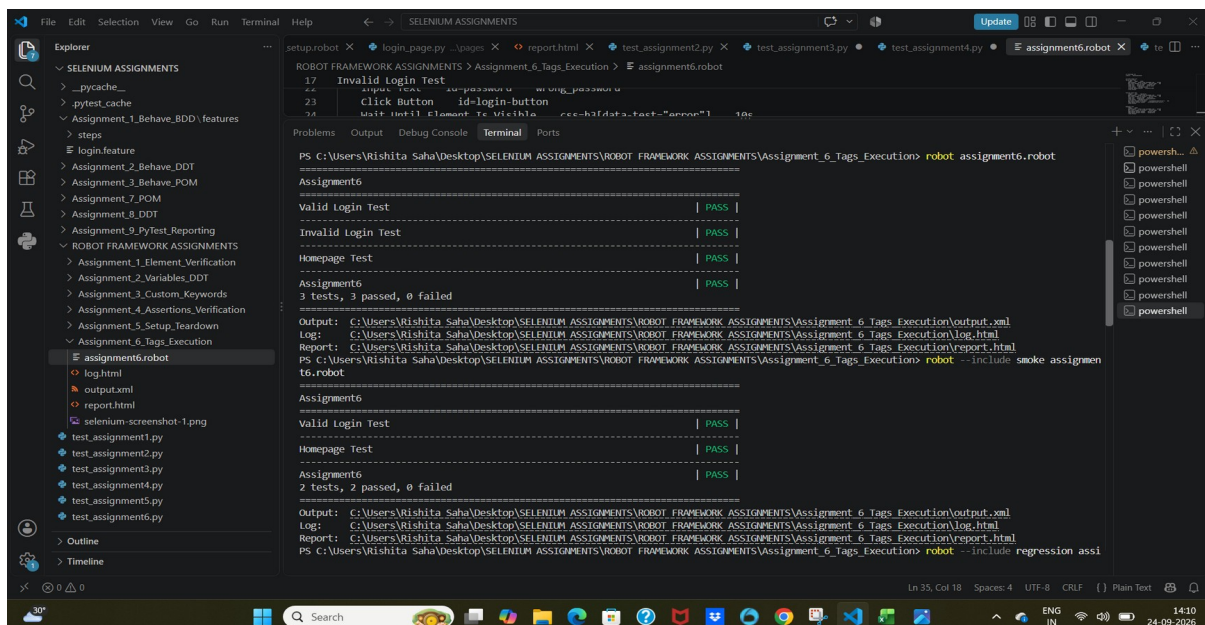
```
pip install robotframework-pabot
```

Run tests in parallel:

```
pabot --processes 2 assignment6.robot
```

6. Output with Clearly Labelled Screenshots

Include the following screenshots in your workbook:

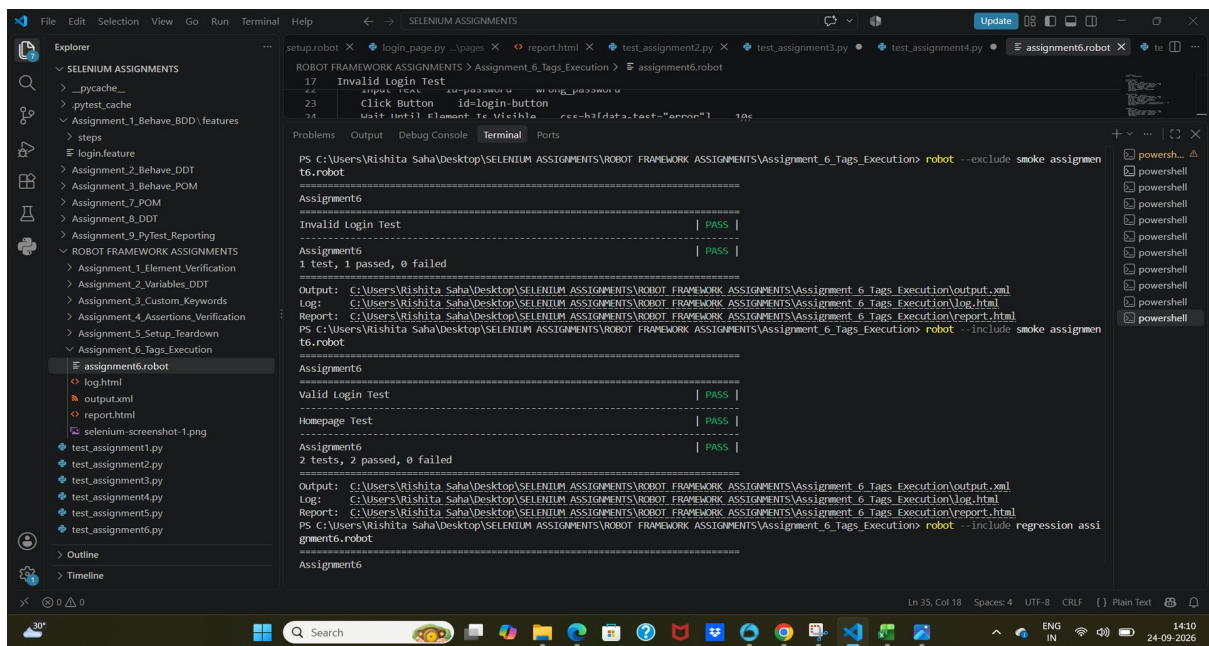


All Test Cases Execution

Show the terminal after:

```
robot assignment6.robot
```

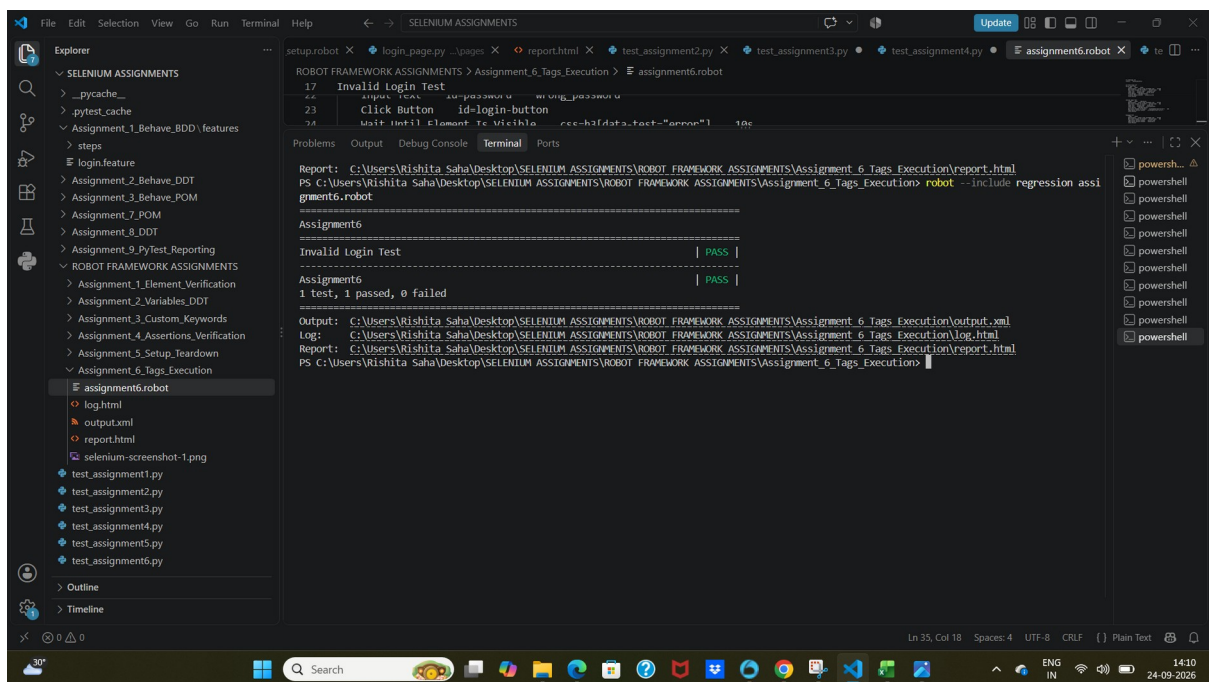
The complete execution should show **3 tests passed**.



Smoke Tag Execution

Show:

robot --include smoke assignment6.robot



Regression Tag Execution

Show:

robot --include regression assignment6.robot

7. Result

The Robot Framework test cases were categorized using **smoke**, **regression**, and **login** tags. The tests can be selectively executed using --include and --exclude, and parallel execution can be performed using Pabot.

The complete test suite successfully contains **3 test cases**.

8. Observation

- Tags allow test cases to be grouped according to their purpose.
- The smoke tag identifies the valid login and homepage tests.
- The regression tag identifies the invalid login test.
- The --include option executes only tests having the specified tag.
- The --exclude option skips tests having the specified tag.
- Pabot allows multiple Robot Framework tests to execute in parallel.

9. Conclusion

This experiment demonstrated **test tagging and selective test execution in Robot Framework**. It also demonstrated how tests can be executed in parallel using **Pabot**, making the test execution process more organized and efficient.